# Gold Prices and Dowry-Related Deaths in India

**Scope:** Three research questions, two primary datasets, and accessible statistical methods. This is an exploratory association study, not a causal analysis.

## 1. Simplified research questions

| Question | Data | Method |
|---|---|---|
| Do annual gold-price changes relate to dowry-death rates in India? | Gold prices + NCRB national figures | Pearson correlation; simple regression |
| Are gold-price changes associated with dowry-death rates 1–2 years later? | Same data, shifted by year | Lagged correlations |
| Does the relationship differ across states? | State-wise NCRB + gold prices | State-wise correlations; plots |

## 2. Datasets (only two required)

**Gold prices:** Annual Indian gold prices (ideally INR per 10 g) from the RBI Database on Indian Economy. Alternative: World Bank Commodity Markets “Pink Sheet” international gold prices, converted to INR if needed. Calculate annual percentage changes. RBI | World Bank.

**Dowry deaths:** Annual state-wise recorded dowry deaths from NCRB *Crime in India* reports. Aggregate carefully for national counts. Use rates per population, preferably relevant female population, when reliable denominators are available. Kaggle mirrors may help with initial processing but should be checked against NCRB. NCRB.

**Not required for the initial project:** NFHS microdata, female literacy, household income, and other socioeconomic control datasets.

## 3. Statistical analysis

**Time-series plots:** Show annual gold-price changes and dowry-death rates. **Correlation:** Calculate Pearson’s r. **Regression:** Estimate the association using a simple model. **Lagged analysis:** Compare same-year, one-year-lag, and two-year-lag results. **State comparison:** Calculate and visualise separate state-wise correlations.

**Simple model:** D(t) = beta0 + beta1 G(t) + error(t), where D(t) is the dowry-death rate and G(t) is the annual percentage change in gold prices.

**One-year lag model:** D(t) = beta0 + beta1 G(t-1) + error(t).

**Interpretation caution:** Correlation does not establish causation. Annual national observations are few; trends, outliers, reporting differences, and shared national gold-price movements can distort results. Check trend sensitivity and avoid treating state-year observations as independent national gold-price shocks.

## 4. Core research papers

| Paper | Role in project |
|---|---|
| Bhalotra, Chakravarty & Gulesci (2020), *The Price of Gold: Dowry and Death in India* | Gold prices, dowry costs, and discrimination against girls (not married-woman dowry deaths). |
| Menon (2020), *The Effect of Marital Endowments on Domestic Violence in India* | Gold prices at marriage and later domestic violence. |
| *Dowry Deaths: A Neglected Public Health Issue in India* (2011) | Background on dowry-death patterns and socioeconomic context. |

Paper links: Bhalotra et al. | Menon | Dowry deaths.

## 5. Workflow and tools

| Stage | Work | Output |
|---|---|---|
| 1. Collect | Download RBI gold prices and NCRB counts | Two source tables |

| Stage | Work | Output |
|---|---|---|
| 2. Prepare | Align years; compute percentage changes and rates | Clean merged CSV |
| 3. Analyse | Plots, correlations, regressions, 1–2 year lags, state comparisons | Figures and statistical results |
| 4. Interpret | Compare with papers; describe uncertainty and limitations | Report and presentation |

**Software:** Python, Pandas, NumPy, Matplotlib/Seaborn, SciPy; optional Statsmodels. No machine learning or advanced panel regression is necessary.

## 6. Final scope recommendation

Keep the three questions above as the core deliverables. Add education or domestic-violence indicators only if the main analysis is complete and the additional data are consistent. Prioritise a reproducible workflow, interpretable graphs, and careful conclusions over more complex models.